PROJECT

STUDY ON THE USE OF FUEL CELLS AND HYDROGEN: HYDROGEN STORAGE SELECTION

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MOTIVATION

- Use of alternative fuels
- Reductions of emissions to air
- It is possible to handle and operate safely hydrogen
- Only the two major problems pending for solutions may make the full application of hydrogen fuel not achievable in the near future: hydrogen storage and production cost.

SUMMARY

- Hydrogen storage is considered to be one of the main obstacles against adopting hydrogen as fuel onboard ships due to its very low energy and due to safety issues. Buy today, some great solutions are developed / developing to use hydrogen onboard.
- The currently available options for storing hydrogen:
  - High Pressure Gas
  - Solid-State Hydrogen Storage
  - Liquid Hydrogen (LH2)
- Currently, there are three common ways of storing hydrogen: as a high-pressure gas, within a compound or chemical host that can store and release hydrogen, or as a cryogenic liquid. Each storage method has different weights and volumes, and for an application such as the Ship or Boot both of these characteristics are critical in understanding whether a storage system is feasible. To aid in evaluation there are two quantities, that can be used to compare the different storage methods.

SUMMARY

- One is the storage system gravimetric specification , defines as: Gravimetric Spec = [Empty Tank Mass (kg)]/[Mass of Stored Hydrogen (kg)].
- An ideal storage system would have a value of zero for the gravimetric spec. Similarly, one can define a volumetric storage specification as: Volumetric Spec = [Outer Tank Volume (L)]/[Mass of Stored Hydrogen (kg)].

- The traditional metal tanks (carbon steel, aluminum) are too heavy for the weight-sensitive high- speed ferry application. Solid-state hydrogen storage is also too heavy and is not yet commercially available. Composite hydrogen tanks are lighter, but are not as gravimetrically or volumetrically attractive as using LH2 storage.

HIGH PRESSURE GAS

- High Pressure Gas

- **Compressed hydrogen in hydrogen tanks under pressure of 350 bar to 700 bar is used for hydrogen tank systems in vehicles. Storing of hydrogen in form of compressed gas is the simplest storage method. It needs a few devices such as a compressor and a pressure tank.**
- **Hydrogen has been stored for decades in carbon steel or aluminum cylinders in applications where weight is not critical such as in laboratory R&D, and in chemical manufacturing.**
- **As the storage pressure increase, capital and operating costs will increase. It is important to know that when compared with traditional fuels, the energy in a compressed hydrogen tank is very low for the same tank volume density.**
- **Providers:**
  - **NPROXX B.V.**
  - **Matheson steel tank**
  - **Lincoln Composites tank**

*** Hydrogen Gas Cylinder**

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SOLID-STATE HYDROGEN STORAGE

- Solid-State Hydrogen Storage

- **The development of hydrogen storage technology has been driven to a large degree by the light-duty fuel cell application, namely, for use in fuel cell passenger cars. This application places greater demands on the "volumetric spec" than on the “gravimetric spec” (although both are important) because of the limited space available on passenger cars.**
- **There are three broad types of hydrogen storage materials that have been investigated: chemical hydrides, sorption materials and metal hydrides.**
- **No material or associated tank system based on advanced complex metal hydrides is close to being commercially available, , requires further development and evaluation.**
- **Providers:**
  - **Ovonic Hydrogen Systems**

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LIQUID HYDROGEN (LH2)

- Liquid Hydrogen

- **LH2 is a cryogenic liquid with boiling point of 20 K (-253.1 °C)**
- **In addition, it can be stored as a constituent in other liquids, such as NaBH4 solutions, rechargeable organic liquids, or anhydrous ammonia NH3. By this method, the weight of hydrogen can be increased by approximately 20 times compared to the compressed form for the same volume.**
- **The tanks which hold it can be thought of as highly engineered Thermos bottles, with an inner metal liner, separated from an outer metal liner with vacuum and perlite insulation (small glass particles) in-between.**
- **Apart from the benefits of light weight and small volume, there are many other attendant benefits of choosing LH2 as the storage method of hydrogen. These benefits include:**
  - **LH2 storage does not require high pressures.**
  - **LH2 storage has been used for decades for space applications and has also been transported on the roads in tankers for decades with an excellent safety record.**

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LIQUID HYDROGEN

- Liquid Hydrogen
  - **With LH2 stored on the vessel, it can in principle be fueled directly from a LH2 tanker brought to the waterfront by the gas supplier.**
  - **LH2 is very similar in its physical and combustion properties to liquid natural gas (LNG)**
  - **LH2 tank technology scales well.**
- **Providers:**
  - **Linde Group**

*** Cryogenic Standard Tanks (LIST2)**

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LIQUID ORGANIC HYDROGEN CARRIER (LOHC) TECHNOLOGY

- LOHC Technology

- **With LOHC Technology, it is possible to store hydrogen in an oil, LOHC enables safe and efficient handling and operations in the existing fuel infrastructure.**
  - **Ideal for small, medium and large-scale hydrogen infrastructure**
  - **Designed for long-life operations and low maintenance**
  - **High safety and high hydrogen storage capacity**
  - **Low footprint and easy on-site installation**

CONCLUSION

- Currently, there are two possible options for storing hydrogen on board, like gas or liquid. For marine use, where large quantities of fuel in addition to long storage time are needed, the storage of hydrogen in the liquid state may appear as the best storage form. LH2 is very similar in its physical and combustion properties to liquid natural gas (LNG) and it is possible to use the LNG infrastructure to handle and operate hydrogen.
- Although liquid hydrogen can provide a lot of advantages, its uses may be restricted because liquefying hydrogen by existing conventional methods consumes a large amount of heat. Practically, compressed hydrogen storage has short filling time and long storage time, while a liquid hydrogen tank has short filling time and short storage time. Moreover, during the long period of storage, to minimize and control the boil-off losses, the permanent cooling using another medium such as liquid nitrogen is needed, which requires additional costs.

CONCLUSION

- In addition, LOHC Technology offers a excellent opportunity to store and release safety hydrogen. This would be a very important requirement, e.g. in the maritime industry.
- The best way of the hydrogen fuel bunkering process will be in the form of liquefied gas. Liquefied hydrogen (LH2) will be stored in cryogenic tanks at the temperature of 20 K and the amount of bunker fuel will depend mainly on the ship's sailing time and engines specific fuel consumption.
- Several issues must be taken into consideration when trying to adopt a new type of fuel like hydrogen, especially for marine applications where strict rules and regulations control the design and manufacture of waterborne vehicles.

REFERENCES

- Feasibility of the SF-BREEZE: a Zero-Emission, Hydrogen Fuel Cell, High-Speed Passenger Ferry, Joseph W. Pratt and Leonard E. Klebanoff, Sept.2016, Sandia National Laboratories
- https://www.the-linde-group.com/en/clean_technology/clean_technology_portfolio/hydrogen_energy_h2/index.html
- https://www.hydrogenious.net/index.php/de/hydrogen-2/